FedDA-HSI: Federated Class-Aware Framework for Hyperspectral Image Classification With Diffusion Augmentation

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***Abstract*—Hyperspectral images (HSIs) have demonstrated remarkable potential in remote sensing applications due to their rich spectral and spatial characteristics. However, the scarcity of labeled HSI data from individual sources significantly hinders the deployment of deep learning models in this domain. Leveraging multisource HSI can mitigate single-source data limitations, but direct data sharing poses serious privacy challenges. Federated learning (FL) provides a compelling framework for privacy-preserving collaborative learning, yet practical challenges remain, particularly data scarcity and the nonindependent and identically distributed (non-IID) nature of client data. In this work, we introduce a diffusion-based data augmentation framework, namely FedDA-HSI, tailored for federated HSI (FedHSI) classification. FedDA-HSI addresses two core issues in real-world FedHSI deployments. First, we propose a federated diffusion model capable of generating high-fidelity synthetic hyperspectral data locally, enabling data augmentation without compromising privacy, while allowing the aggregated global model to capture shared knowledge across clients. Second, we incorporate a class-aware augmentation strategy to alleviate data imbalance by injecting synthetic samples of missing or underrepresented classes, effectively improving interclient data diversity and mitigating non-IID effects. Extensive experiments on benchmark HSI datasets validate that our approach significantly improves classification performance under heterogeneous conditions.**

***Index Terms*—Data heterogeneity, diffusion model, federated learning (FL), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture rich spatial and continuous spectral information, making them valuable in diverse applications such as mineral exploration [1], environmental monitoring [2], [3], and military reconnaissance [4]. Deep learning techniques have shown significant promise in HSI analysis, driving notable advancements in recent years [5], [6]. However, several key challenges continue to hinder the effective deployment of deep models in this domain. These include the limited availability of labeled HSI datasets [7], [8], the substantial computational and memory requirements associated with processing high-dimensional spectral data [9], [10], [11], and growing privacy concerns surrounding the handling of sensitive geospatial information [12]. Collectively, these factors limit the scalability and real-world applicability of traditional deep learning approaches in HSI [13], [14].

Federated learning (FL) has emerged as a promising paradigm to address these limitations. In real-world scenarios, HSIs are often distributed across multiple sources or institutions, where each client holds limited and partially distinct samples. By decentralizing the training process, FL enables collaborative model development across multiple clients while keeping data local. This not only alleviates the computational burden on individual participants but also significantly reduces privacy risks [15], [16], [17]. As a result, FL facilitates knowledge sharing across data silos without compromising sensitive information, offering a compelling solution for hyperspectral learning.

Although FL eases the computational burden and protects data privacy by decentralizing training [15], [18], [19], it does not solve the fundamental scarcity of labeled hyperspectral data. Annotating remote sensing imagery remains costly and labor-intensive. In addition, FL introduces a new obstacle: pronounced data heterogeneity resulting from nonindependent and identically distributed (non-IID) client datasets [18], [20], [21]. In hyperspectral scenarios, this heterogeneity appears in two ways. First, data volume is uneven because satellites or aerial platforms capture scenes of very different sizes, so some clients possess abundant samples while others hold only a few. Second, class distributions are highly skewed because land-cover types vary geographically, and certain clients may observe many instances of a particular class while others see none [22]. These imbalances cause local models to overfit the dominant classes in their own data and to follow divergent optimization trajectories. When such misaligned gradients are aggregated, the global model drifts and struggles to converge [23]. Enabling robust learning with limited annotations while also mitigating the adverse effects of non-IID data, all without

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relaxing privacy guarantees, therefore, remains a central open challenge in federated HSI (FedHSI) research.

To alleviate data scarcity, data augmentation is widely used to boost model generalization [24], [25]. Most hyperspectral synthesis methods [11], [26] rely on centralized architectures, which do not account for data heterogeneity across multiple sources, privacy concerns, or the computational limitations of edge devices. Motivated by these challenges, recent works have begun integrating generative models with FL, applying variational autoencoders (VAEs), generative adversarial networks (GANs), and particularly diffusion models to enlarge local datasets while protecting privacy [27], [28], [29], [30], [31], [32]. However, existing federated generation methods have mostly focused on RGB images [34]. HSI generation involves significantly higher data dimensionality and requires capturing much richer spectral information, making it substantially more challenging. As a result, HSI generation remains largely unexplored in federated settings. Diffusion frameworks are attractive because of their stability and high-fidelity outputs. We adopt a federated diffusion approach for HSI and introduce spectral unmixing to reduce dimensionality while retaining critical spatial and spectral detail [35], [36], [37]. In this way, each client can augment its data, share only model parameters, and achieve performance gains even with limited labels.

To ease the impact of data heterogeneity in FL, researchers have explored two main directions. Uneven data volumes are typically handled by weighting client updates by sample size, ensuring fairer aggregation [18]. Imbalanced label distributions are often addressed by supplementing training with large public datasets [38], [39], [40], yet such data are rarely available in practice. Data augmentation can ease data scarcity for clients and partially mitigate uneven data volumes. However, existing HSI augmentation methods are class-agnostic and ineffective against local label imbalance. We tackle this gap with a class-aware augmentation scheme built on federated conditional diffusion models. Each client synthesizes samples for missing or rare classes, enriching local data while keeping raw observations private. The resulting balanced label distribution stabilizes model training and accelerates convergence, leading to improved hyperspectral classification accuracy under realistic heterogeneous conditions.

We present FedDA-HSI, a class-aware, diffusion-based augmentation framework that tackles both data scarcity and heterogeneity in FedHSI classification. Each client first trains a diffusion model locally, learning to generate synthetic hyperspectral samples that mirror its private distribution. Periodic federated aggregation merges these local models into a global diffuser, sharing knowledge without exposing raw data. The global model is then redeployed to every client, where it produces additional, high-fidelity HSI samples, expanding local training sets while safeguarding privacy. To further mitigate the impact of non-IID data across clients, we add a class-aware strategy that synthesizes data only for missing or rare classes, equalizing class distributions and boosting robustness. Our main contributions are as follows.

1) We introduce FedDA-HSI, a federated diffusion framework that augments hyperspectral data entirely on-device. Each client produces high-fidelity synthetic samples that expand its local training set while keeping raw imagery private, yet the aggregated model still captures global knowledge.
2) To counter non-IID effects, we design a class-aware augmentation mechanism that synthesizes data only for missing or underrepresented classes. Balancing label distributions across clients increases data diversity and markedly boosts classification accuracy and robustness.
3) Across multiple benchmark datasets, the proposed framework consistently outperforms leading federated baselines on HSI classification tasks.

II. Related Works

A. HSI Processing With Deep Learning

Deep learning has played a crucial role in advancing HSI processing tasks such as denoising, anomaly detection, and classification. For instance, HyADD [41] proposes a hierarchical framework for simultaneous anomaly detection and denoising in HSIs by iteratively coupling the two tasks. Leveraging spatial–spectral gradients and low-rank subspace priors, HyADD effectively enhances robustness against real-world noise while preserving critical structural and semantic information. In the context of classification, NSCKL [42] introduces a normalized spectral clustering scheme under manifold assumptions for semi-supervised HSI classification. By combining local and global spatial correlations through graph construction and kernel-based iterative filtering, NSCKL generates clustered features that enhance classification performance in noncontiguous regions. These works reflect recent trends in deep learning-based hyperspectral representation learning and provide a valuable reference background for our study.

B. HSI Augmentation

Hyperspectral data augmentation is challenging because of its high dimensionality and complex spectral structure. Early work relied on hand-crafted spectral manipulations, such as mixing spectra within a class or adding small perturbations, to increase sample diversity [43]. These techniques, however, adapt poorly to the nonlinear variability of real scenes. Data-driven generative models promise greater flexibility by learning spectral distributions directly from data. Yet conventional choices, such as VAEs [30] and GANs [31], translate imperfectly to hyperspectral imagery: VAEs often produce spectrally blurry outputs [44], and GANs suffer from training instability, mode collapse, and limited spectral fidelity [45]. These issues become more severe when dealing with high-dimensional hyperspectral data [46], [47]. In contrast, diffusion models provide a more stable and expressive alternative. They reduce mode collapse, better capture complex spectral distributions, and more effectively preserve interband spectral correlations [36], [48].

To manage the large spectral dimension of HSIs, recent diffusion-based methods embed dimensionality-reduction steps. R2H-CCD synthesizes full HSIs from RGB inputs

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via conditional cascade diffusion [49], and VQ-GANs compress spectra into latent codes before diffusion [50]. Because these approaches ignore physical constraints, newer work first performs spectral unmixing by decomposing an HSI into endmember signatures and abundance maps, then applies diffusion in the low-dimensional abundance space. Examples include RGB-guided endmember and abundance fusion [37] and the UnmixDiff framework, which generates HSIs directly from abundance maps [36]. This physics-aware strategy preserves spectral consistency while reducing computational load.

Despite these advances, centralized generative pipelines still demand substantial memory and direct access to raw hyperspectral data, raising privacy concerns for sensitive geospatial content. These drawbacks motivate an FL paradigm for HSI augmentation, enabling collaborative model training without sharing raw imagery.

C. FL With Non-IID Data

FL, first introduced by Google [51], [52], enables collaborative model training while keeping data local, representing a clear advance for privacy-preserving analytics. Yet when client data are highly non-IID, FL can converge slowly and yield suboptimal accuracy. Researchers have tackled this challenge along several lines. One strand relies on limited data sharing; for example, HHH-FL and FedDG exchange a subset of features, whereas FedMD and ensemble-based schemes share a small public dataset, but such strategies erode privacy guarantees [53], [54], [55], [56]. A second line introduces update-level regularization: FedProx stabilizes local optimization with a proximal term, and hybrid networks split global backbone layers from locally personalized heads [57], [58], [59]. A third, representation-focused track uses conditional VAEs, adaptive local aggregation (FedALA), or feature and classifier decoupling (FedPAC) to align client updates without extra communication cost, achieving strong results on non-IID benchmarks [27], [60], [61]. Despite this progress, nearly all work assumes conventional RGB or tabular data. Hyperspectral sensing introduces a different order of heterogeneity: clients image distinct landscapes, producing highly skewed label distributions and divergent spectral signatures, all embedded in hundreds of bands. This combination of spectral complexity, label imbalance, and high dimensionality pushes existing FL algorithms beyond their design scope, exposing an urgent research gap in hyperspectral analytics.

D. HSI Classification

HSI classification has received continuous attention in remote sensing due to its ability to capture detailed spectral signatures across numerous contiguous bands. Recent advancements have focused on improving classification performance under challenging scenarios such as limited supervision, open-set recognition, and multimodal data fusion.

For example, ReSC-Net [62] improves small-sample classification by combining online spectral augmentation with a lightweight residual attention architecture. HyperTaFOR [63] and HyLiOSR [64] target open-set recognition using transformer- and autoencoder-based frameworks, respectively. MMD-MLP [65] integrates multiresolution spectral features and LiDAR data through a cross-modal attention mechanism for enhanced land-cover mapping. SSEFN [66] introduces a spatial–spectral enhancement strategy with adaptive fusion to address the few-shot learning setting. More recently, HyperSigma [67] proposes a scalable vision transformer-based foundation model pretrained on a large-scale HSI dataset, aiming at generalizable HSI interpretation across diverse tasks and domains.

While these centralized methods achieve strong performance, they assume access to fully labeled and centrally stored datasets, which may not be feasible in real-world applications. In contrast, our proposed method addresses FedHSI classification, where data privacy, label scarcity, and non-IID client distributions are tackled through a novel diffusion-based data augmentation strategy and class-aware sample balancing.

E. FedHSI Classification Under Non-IID Scenarios

Recent efforts have begun to explore FL in remote sensing scenarios under non-IID conditions. FedFusion [68] proposes a manifold-driven FL framework for multisatellite and multimodal image fusion by learning low-rank representations across clients, showing promising results under heterogeneous data. Transformer-based FL methods [69] investigate the robustness of lightweight attention architectures for multilabel classification tasks in remote sensing, and report improved adaptability to data distribution shifts. In addition, a comprehensive benchmark study [70] systematically compares FL algorithms for remote sensing image classification, emphasizing the challenges brought by label imbalance and decentralized archives.

However, these approaches do not effectively address the challenges of limited hyperspectral data and label imbalance. To this end, we propose FedDA-HSI, a diffusion-based augmentation framework that generates class-aware synthetic samples to locally enrich underrepresented categories. This generative mechanism helps reduce label shift across clients and improves classification performance under practical non-IID settings.

III. Methodology

A. Problem Definition

In the FL-based HSI classification task, there is a server and K clients (indexed by k). Each client naturally holds a local private dataset composed of HSI patches of fixed size (height H, width W, and number of bands B), reflecting the unique data distribution of its sensing region. The local dataset is denoted as $D_k = \{(x_i^k, y_i^k)\}_{i=1}^{N_k}$, where $x_i^k \in \mathbb{R}^{B\times H\times W}$ represents a single HSI patch, y_i^k denotes its corresponding label, and N_k is the number of samples in the dataset of client k. The server aims to guide clients in collaboratively learning a global model with parameters θ. The federated optimization objective is formulated as

$$\min_{\theta} f(\theta) = \sum_{k=1}^{K} \frac{|D_k|}{|D|} f_k(\theta) \tag{1}$$

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Algorithm 1 Federated Diffusion Model

Input: The set of clients C, local dataset D_k, local labels y_k, all datasets D, learning rate η, number of diffusion timesteps T, variance schedule $\beta_1, \ldots, \beta_T$, number of local epochs T_c, communication rounds R, local diffusion model θ_r^k.
Output: Global diffusion model θ_r
procedure Server
  initialize θ_0
  for $r = 1, \ldots, R$ **do**
    receives θ_r^k from all clients
    $\theta_r \leftarrow \sum_{k=1}^{K} \frac{|D_k|}{|D|} \cdot \theta_r^k$
    sends θ_r to all clients
  end for
  return $\theta_{r=R}$
procedure Client θ_r^k
  for each client $k \in C$ **in parallel do**
    $\theta_r^k \leftarrow \theta_{r-1}$
    for $t = 1, \ldots, T_c$ **do**
      for mini-batch $b \in D_k$ **do**
        $\theta_r^k \leftarrow \theta_r^k - \eta \nabla_{\theta_r^k} \mathcal{L}_{\text{diff}}(b, \theta_r^k)$
      end for
    end for
  end for
  sends θ_r^k to the server
procedure $\mathcal{L}_{\text{diff}}(b, \theta_r^k)$
  for $i \in b$ **do**
    $t \sim \text{Uniform}(1, \cdots, T)$
    $\epsilon_t \sim \mathcal{N}(0, I)$
    $\bar{\alpha}_t = \prod_{s=1}^{t}(1 - \beta_s)$
    $\mathcal{L}_i = \|\epsilon_t - \epsilon_{\theta_r^k}(\sqrt{\bar{\alpha}_t} x_i + \sqrt{1 - \bar{\alpha}_t}\epsilon_t, t, y_k)\|_2^2$
end for
return $\frac{1}{|b|} \sum_{i \in b} \mathcal{L}_i$

Algorithm 2 Federated Hyperspectral Classification

Input: The set of clients C, local dataset D_k, local labels y_k, all datasets D, learning rate η, number of local epochs T_c, number of communication rounds R, local parameters θ_r^k, global conditional diffusion model $\mathcal{G}$.
Output: Global classification model θ_r
procedure Server
  Initialize θ_0
  for $r = 1, \cdots, R$ **do**
    Call the client procedure
    $\theta_r \leftarrow \sum_{k=1}^{K} \frac{|D_k|}{|D|} \cdot \theta_r^k$
    Send θ_r to all clients
  end for
  return $\theta_{r=R}$
procedure Client
  for each client $k \in C$ **in parallel do**
    Generate abundance maps $A_{syn}^k = \mathcal{G}(\epsilon, y_k)$
    Decode abundance maps into synthetic HSI $H_{syn}^k = \theta_D^k(A_{syn}^k)$
    Augment local dataset with synthetic data $D_{aug}^k = D_k \cup H_{\text{syn}}^k$
    Initialize with global model from server: $\theta_r^k \leftarrow \theta_{r-1}$
    for $t = 1, \cdots, T_c$ **do**
      for mini-batch $b \in D_{aug}^k$ **do**
        $\theta_r^k \leftarrow \theta_r^k - \eta \cdot \nabla \mathcal{L}_c(b, \theta_r^k)$
      end for
    end for
  end for
  Each client sends θ_r^k to the server

where $f_k(\theta)$ denotes the loss function over a local client dataset $D_k \subset D$ under global model parameter θ. D represents the union of all client datasets in FL. The contribution of each client k to the global objective is weighted by $|D_k|/|D|$, proportional to its dataset size. In the FL network, all clients maintain direct connections with the server but not with each other. Clients and the server collaboratively perform different parts of the training process. Clients perform local model training while the server manages global model aggregation through parameter exchange. This distributed architecture leverages client-side resources to reduce the computational load on the central server. Furthermore, FL enables collaborative model training while keeping sensitive data localized at each client, ensuring data privacy.

However, two critical challenges emerge in practical FL deployment. First, the inherent scarcity of labeled data in hyperspectral imaging poses a significant limitation, as obtaining accurate pixel-wise annotations for HSI data is both time-consuming and labor-intensive. Second, the non-IID nature of client data presents a fundamental challenge in FL. In land cover classification, this heterogeneity primarily manifests as an uneven distribution of classes across clients, as each client's data typically originates from distinct geographical regions with varying land cover compositions. This statistical divergence can lead to biased local models and suboptimal global model convergence.

B. Overall Architecture

We propose FedDA-HSI, a FL framework for HSI classification that addresses both data scarcity and non-IID challenges through diffusion-based data augmentation. The framework is illustrated in Fig. 1.

FedDA-HSI operates in two stages. In the first stage, each client trains a local generative model consisting of an unmixing model and a diffusion model using their private HSI data. To preserve privacy while enabling collaborative learning, only the diffusion model parameters are shared with the central server for federated aggregation across multiple rounds, resulting in a robust global generative model. In the second stage, we introduce a class-aware data augmentation strategy where clients utilize the global diffusion model to generate synthetic samples for missing or underrepresented classes. The process involves generating abundance maps through the diffusion model, which are then transformed into realistic HSI data via the unmixing decoder. This targeted augmentation approach helps balance the class distribution across clients. Finally, each client combines its original data with the synthetically generated samples to create an enhanced, more diverse training dataset. The classifier is then trained using standard FL procedures, where the balanced datasets help mitigate the negative

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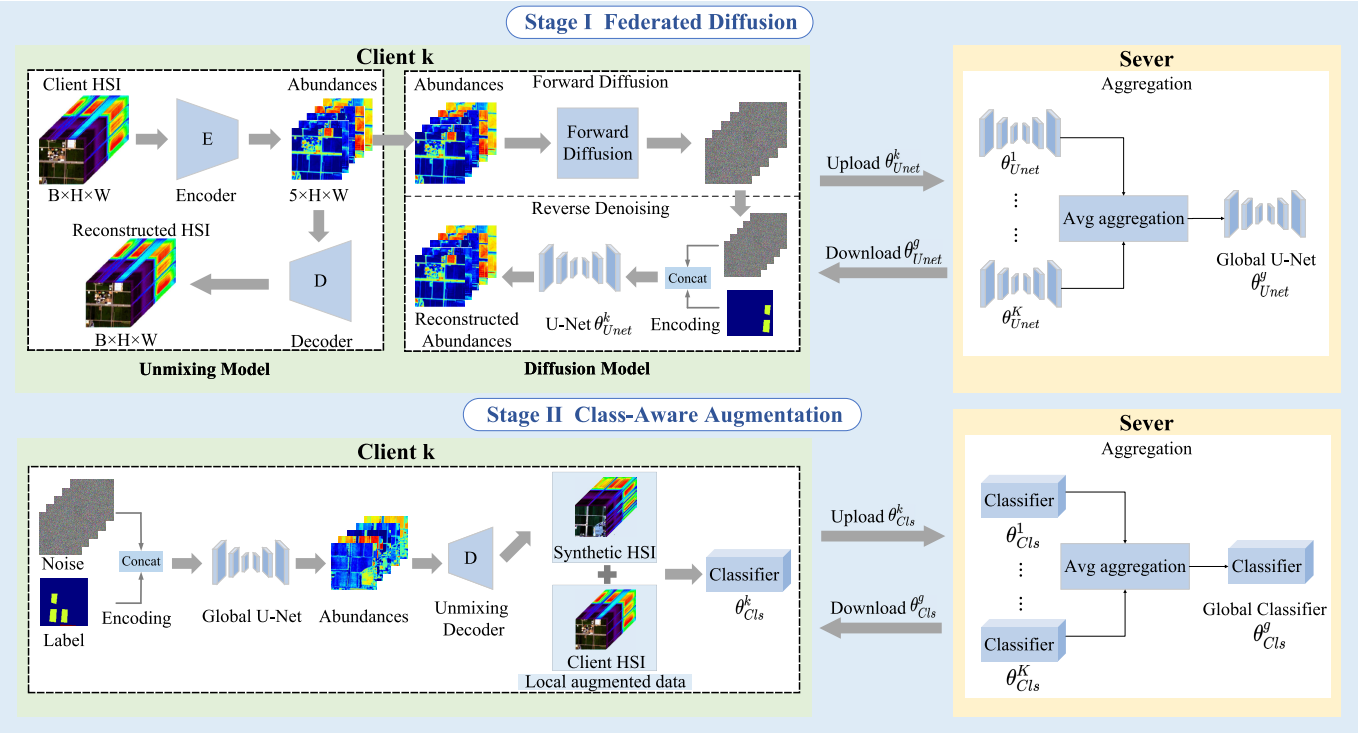


Fig. 1. Overall framework of FedDA-HSI. The pipeline involves two stages. *Stage I:* Clients train local generative models and upload diffusion parameters for multiround federated aggregation. *Stage II:* Clients input missing or underrepresented labels into the global diffusion model to generate abundance maps, which are then decoded by the unmixing decoder to synthesize HSIs. Subsequently, original and synthetic data are combined to train a federated classifier.

effects of data heterogeneity while maintaining data privacy throughout the process. FedDA-HSI effectively addresses the dual challenges of data scarcity and non-IID data in FedHSI classification through synthetic data generation and strategic augmentation, leading to improved classification performance under heterogeneous conditions. The detailed training process is outlined in Algorithm 1 and Algorithm 2.

C. Federated Diffusion Model

In the first stage, we aim to train a federated diffusion model for synthetic HSI generation. The design of the federated diffusion model must satisfy two core requirements. The first requirement is privacy protection, which ensures that each client's original data is always kept locally. Only indirect knowledge transfer occurs through model parameters. The second requirement is to guarantee the quality of the generated output. When working with non-IID data, it is essential to maintain both the quality and diversity of the generated samples within the expected range.

To meet these requirements, we develop a framework that combines spectral unmixing with conditional diffusion models under FL. Due to the high dimensionality of hyperspectral data, the data first undergoes dimensionality reduction through the unmixing model to preserve key structural and spectral information. The unmixing model is based on a linear mixing model [35], representing the HSI as a linear combination of end members and abundance maps. It can be expressed as follows:

$$X = EA + N \tag{2}$$

where $X \in \mathbb{R}^{B\times H\times W}$ is the HSI with B bands, H height, and W width. $E \in \mathbb{R}^{B\times p}$ represents the endmember matrix with p endmembers. $A \in \mathbb{R}^{p\times H\times W}$ indicates the abundance maps. $N \in \mathbb{R}^{B\times H\times W}$ is the bias item representing natural noise. Notably, the number of endmembers p is significantly smaller than the number of spectral bands B, enabling effective dimensionality reduction.

The unmixing model uses an autoencoder structure. Specifically, every client $k \in C$ trains an unmixing autoencoder to convergence on its private local dataset, where C denotes the set of all participating clients in FL. The encoder θ_E maps the original HSI to abundance maps, while the decoder θ_D reconstructs the HSI. The decoder functions as a linear mixture model, with its weights representing the extracted end members. Given an abundance map A as input, the decoder linearly combines the endmembers to reconstruct the HSI $\hat{X}$. The training objective is to minimize the reconstruction error between the original image X and the reconstructed image $\hat{X}$. This process can be formulated as

$$\hat{X} = \theta_D\left(\theta_E\left(X\right)\right) = \theta_D\left(A\right). \tag{3}$$

After training the spectral unmixing autoencoder locally on each client, the encoder is used to extract abundance maps from the original HSI data, providing compact, low-dimensional representations while preserving meaningful spectral and spatial information. To facilitate the reconstruction of synthetic HSIs in later stages, each client uploads its trained decoder, which contains the estimated endmembers, to the central server for shared use. In this way, the generation

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task is decoupled. Clients are only responsible for synthesizing abundance maps under privacy constraints, while the decoders shared by the server ensure consistent reconstruction capability across clients.

With the abundance maps extracted and decoder parameters shared, the generative modeling task is now shifted to the abundance space, allowing the use of lightweight conditional diffusion models to synthesize class-consistent abundance maps under privacy-preserving constraints. This effectively bridges the high-dimensional hyperspectral input and generative model training in a fully federated setting.

The synthetic task is projected from the hyperspectral space to the abundance space through the unmixing process [36]. The conditional diffusion model learns to generate abundance maps by gradually denoising Gaussian noise through a series of steps with the condition of semantic segmentation maps y. The diffusion model consists of two stages: forward diffusion and reverse denoising [32]. In the forward process, Gaussian noise is gradually added to the abundance map, resulting in an image that is close to pure noise. In the reverse process, a neural network is used to learn the denoising procedure, gradually recovering the abundance map from pure noise conditioned on y. The noise prediction network employs a UNet structure, aiming to minimize the error between the predicted noise and the actual noise. The synthetic target can be expressed as follows:

$$\hat{A} = \mathcal{G}(z, y), \quad \hat{A} \sim A \tag{4}$$

where $\mathcal{G}$ is the trained diffusion model, y is the label map, and z is the input random noise. Given an input abundance map A, the model predicts a denoised abundance map $\hat{A}$. $\hat{A} \sim A$ indicates that $\hat{A}$ follows the same distribution as A.

For each client k, the local conditional diffusion model is optimized by minimizing the local diffusion loss, defined as

$$\mathcal{L}_{\text{diff}}^{(k)}\left(\theta_r^k\right) = \mathbb{E}_{t,\epsilon_t}\left[\left\|\epsilon_t - \epsilon_{\theta_r^k}\left(\sqrt{\bar{\alpha}_t}A_0 + \sqrt{1-\bar{\alpha}_t}\epsilon_t, t, y\right)\right\|_2^2\right] \tag{5}$$

where A_0 is the original abundance map, y is the corresponding segmentation label, t is sampled uniformly over diffusion steps, $\epsilon_t \sim \mathcal{N}(0, I)$ is Gaussian noise, and $\epsilon_{\theta_r^k}$ is the noise estimator (UNet) of client k. The client updates its model parameters θ_r^k by minimizing this loss via mini-batch SGD.

In the FL framework, the training process is designed as an iterative collaboration between K distributed clients and a central server. Its goal is to jointly train a global diffusion model without sharing raw data, thus preserving local data privacy while leveraging the collective knowledge of all participants. To begin, each client locally trains its own diffusion model using its private abundance maps. Specifically, at each round r, the client optimizes its local denoising network parameters θ_r^k over T_c local epochs using mini-batch stochastic gradient descent. This training minimizes a diffusion loss function defined only on the client's local data. Once local training is completed, the updated model parameters θ_r^k are sent to the central server. The server then aggregates the parameters from all clients to form a new global model θ_r. This is achieved via the federated averaging (FedAvg) algorithm, where the contribution of each client is weighted proportionally to the size of its local dataset

$$\theta_r = \sum_{k=1}^{K} \frac{|D_k|}{|D|} \cdot \theta_r^k \tag{6}$$

where θ_r represents the aggregated global diffusion model at round r, θ_r^k denotes the local diffusion model parameters uploaded by client k, and $|D_k|/|D|$ is the weight assigned to each client based on its local dataset size, with $|D|$ being the total size of all datasets and $|D_k|$ being the size of dataset on client k. After aggregation, the server broadcasts the updated global parameters θ_r back to all clients. Each client then uses these parameters to initialize its model before starting the next round of local training. This federated optimization process continues for a total of R communication rounds. Through this cycle of local training and global aggregation, the model gradually learns to generate high-quality abundance maps across diverse data distributions, without requiring any centralized access to client data. The overall training objective of the proposed FedHSI-Diff framework is to minimize the weighted sum of local diffusion losses across clients

$$\min_{\theta} \sum_{k=1}^{K} \frac{|D_k|}{|D|} \cdot \mathcal{L}_{\text{diff}}^{(k)}(\theta) \tag{7}$$

where θ denotes the global denoising network shared across clients, and $\mathcal{L}_{\text{diff}}^{(k)}(\theta)$ represents the local diffusion loss computed on the abundance maps in client k. This concise formulation captures the goal of collaboratively optimizing the generative model. The detailed procedure is presented in Algorithm 1.

D. Class-Aware Augmentation Strategy

In the second stage, we propose a novel class-aware augmentation strategy that harnesses our trained federated diffusion model to tackle the challenge of data heterogeneity. Building upon this augmented data, we train a federated classifier to enhance the overall classification performance.

The global diffusion model serves as a powerful tool for knowledge integration, effectively capturing the diverse data characteristics from all clients through model aggregation. This enables the generation of high-quality synthetic abundance maps that naturally align with each client's data distribution. The generation process begins with sampling Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, which is combined with target class labels y_k. Through the reverse diffusion process, the model generates synthetic abundance maps, transforming the initial random noise into meaningful representations. This process can be formulated as

$$A_{\text{syn}}^k = \mathcal{G}(\epsilon, y_k; \theta_R) \tag{8}$$

where θ_R denotes the parameters of the trained global diffusion model, y_k represents the input label of the missing class on client k, ϵ is the random noise, and A_{syn}^k represents the synthetic abundance map generated on client k. This approach ensures the preservation of essential spectral characteristics. The synthetic abundance maps are then transformed into HSIs

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using client-specific decoder parameters. This process can be formulated as

$$H_{\text{syn}}^{k} = \theta_{D}^{k}\left(A_{\text{syn}}^{k}\right) \tag{9}$$

where θ_D^k represents the trained decoder parameters on client k, which transforms the synthetic abundance maps A_{syn}^k into synthetic HSIs H_{syn}^k through the unmixing decoder. Notably, this framework maintains strict data privacy by sharing only model parameters rather than raw data, while simultaneously enabling the creation of balanced class distributions across all clients.

The training process intelligently combines original and synthetic data into an augmented training set $D_{\text{aug}}^k = D_k \cup H_{\text{syn}}^k$, enhancing data diversity while preserving local data characteristics. Each client then trains its local classifier by minimizing the cross-entropy loss function using stochastic gradient descent. The federation process coordinates client collaboration through the FedAvg algorithm, aggregating local classifier parameters at the server. This aggregated global classifier is distributed back to participating clients, initiating an iterative process that continues for a predetermined number of rounds. Throughout this cycle, the model's generalization capabilities continuously improve through effective knowledge sharing, while the augmented dataset helps overcome data heterogeneity challenges. This integrated approach creates a robust learning framework that simultaneously addresses class imbalance and enhances model performance, while maintaining the privacy guarantees in FL architectures. The detailed procedure is presented in Algorithm 2.

E. Privacy Protection Mechanism

Privacy preservation is a critical concern in FedHSI analysis, especially when training models across clients with sensitive data. Our FedDA-HSI framework incorporates multiple privacy-preserving strategies to ensure client data confidentiality while maintaining model performance.

1) FL Foundation: The foundation of our approach lies in FL, where all raw hyperspectral data remain strictly local to each client. The global server coordinates training via model aggregation rather than data centralization, thereby inherently safeguarding raw data from exposure or sharing [71], [72], [73].

2) Generator-Centric Aggregation: Unlike conventional FL systems that aggregate classification models, FedDA-HSI federatively trains *generative diffusion models* on each client. This key design choice improves privacy resilience, as most known threats, such as model inversion, feature leakage, or gradient inference, are primarily targeted at discriminative classifiers [74], [75]. In contrast, diffusion-based generators learn the data distribution indirectly through denoising objectives, making individual data reconstruction significantly more difficult.

3) DP During Local Generation: To further enhance privacy, we integrate a lightweight differential privacy (DP) mechanism into the on-device data generation phase [76], [77], [78]. Specifically, we inject calibrated Gaussian noise into the generated abundance maps. This ensures that the resulting synthetic samples deviate from the exact local distribution, preventing distribution memorization or instance-level leakage, while still maintaining semantic fidelity for data augmentation.

4) Experimental Validation: In Section IV, we quantify the tradeoff between privacy and utility by comparing the performance of DP-enhanced generation against the standard setting. The results confirm that our DP strategy introduces minimal degradation in classification accuracy while significantly increasing robustness against potential data reconstruction threats.

IV. Experiments

A. Experimental Settings

1) Datasets: We employed four representative hyperspectral datasets to evaluate the generalization capability of our proposed method. The Chikusei dataset[1] It [79] comprises 128 spectral bands spanning from 363 to 1018 nm, with a spatial dimension of 2517×2335 pixels. The Berlin-Urban-Gradient dataset[2] is acquired from satellite imagery covering Berlin's urban to rural transition zones and is accessible for academic use. This dataset [80] has a spatial resolution of 2722×732 pixels and includes 111 spectral bands with a wavelength range of 450–2450 nm. The Xiongan dataset[3] is collected via UAV imaging over the Xiongan New Area in China as part of an institutional collaboration. It [81] contains 3750×1580 pixels and 250 spectral bands, covering a wavelength range from 400 to 1000 nm. It includes 19 land cover types, such as water, vegetation, and buildings. The Houston 2018 dataset[4] is provided by the IEEE GRSS Data Fusion Contest and is freely available online for public research. This dataset [82] has a spatial resolution of 601×2384 pixels and includes 48 spectral bands with a wavelength range of 380–1050 nm. This dataset consists of 20 categories and covers the university campus and its surrounding areas. Our experimental framework was designed to evaluate both the classification performance and the generative capabilities of the proposed approach. We first conducted classification experiments on the Xiongan and Houston 2018 datasets to validate the effectiveness of the proposed class-aware data augmentation strategy in addressing the challenges posed by heterogeneous hyperspectral data distributions in FL. In addition, we assessed the performance of the federated diffusion model for HSI synthesis. Specifically, the Chikusei and Berlin-Urban-Gradient datasets were used for unconditional generation, while the Xiongan and Houston 2018 datasets were used for conditional generation tasks. These experiments are designed to examine the effectiveness of federated training of the diffusion model in facilitating knowledge transfer across clients, while maintaining strict data privacy at the client level.

2) Data Splitting: To ensure consistent preprocessing across all datasets, we first cropped the original HSIs into patches of size 128×128 pixels with a 40% spatial overlap,

[1]https://eod-grss-ieee.com/dataset-detail/Q0d5eVVKcWNPZzY0WTlWRmk2c2xoZz09 is an airborne dataset collected over Chikusei city in Japan and is publicly available for research use.

[2]https://dataservices.gfz-potsdam.de/enmap/showshort.php?id=escidoc:1823890

[3]https://www.ygxb.ac.cn/zh/article/doi/10.11834/jrs.20209065/

[4]IEEE GRSS 2018 Data Fusion Challenge.

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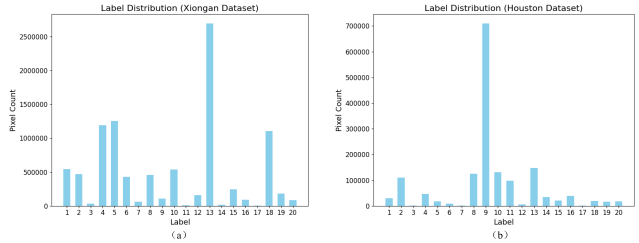


Fig. 2. Class distribution of the Xiongan and Houston 2018 datasets. (a) Xiongan dataset. (b) Houston 2018 dataset.

thereby increasing the number of training samples while preserving local spectral–spatial structures. These patches were then randomly distributed among clients to simulate an FL scenario. Each client received an equal number of patches. However, due to the inherent spectral and spatial variability within HSIs, the resulting class label distributions across different clients remained imbalanced, naturally leading to a non-IID data partitioning. After preprocessing, the number of resulting image patches for each dataset is as follows: the Chikusei dataset yields a total of 837 patches, the Berlin-Urban-Gradient dataset contains 300 patches, the Xiongan dataset provides 800 patches, and the Houston 2018 dataset results in 280 patches. We also visualize the class distribution for the Xiongan and Houston 2018 datasets, as shown in Fig. 2. These distributions highlight intrinsic imbalances within real-world scenes, which pose unique challenges to federated hyperspectral classification and further motivate our class-aware generation strategy.

This client-level patch assignment was used for training the diffusion model in a federated manner. For the downstream classification task, each client retained its originally assigned subset of patches to maintain consistency with the prior federated training setting. Pixel-wise classification was performed using 7×7 spatial neighborhoods centered around each target pixel, allowing the model to exploit both spectral and local contextual information. Within each client, the patch-level data were further split into 40% for training, 30% for validation, and 30% for testing. This intraclient split ensures that each client performs local training and evaluation independently, emulating a realistic decentralized learning environment. For the final evaluation, we aggregated all local client test sets to construct a unified global test set, which provides a comprehensive measurement of the model's generalization ability across heterogeneous data sources. To further investigate the impact of data heterogeneity and client imbalance, we also conducted parameter sensitivity experiments using Dirichlet distribution-based data partitioning under varying concentration parameters, as detailed in the subsequent sections.

3) Implementation Details: All experiments were implemented using the PyTorch framework. For the unmixing model configuration, we set the number of endmembers to 5 and conducted training over 80 epochs. The optimization was performed using Adam optimizer with an initial learning rate of $2e^{-4}$, complemented by a stepped learning rate decay schedule for dynamic adjustment. In the diffusion model, we set the time step T to 2000. The diffusion coefficients β_1 and β_T are set to $1e^{-6}$ and $1e^{-2}$, implementing a linear diffusion process. The training phase employed the Adam optimizer with an initial learning rate of $1e^{-4}$. The model was trained for 500 000 local epochs across four communication rounds. For classification, we implemented a 3-D convolutional neural network architecture. The model processes input patches of size 7, with training conducted over two local epochs and four communication rounds. All experiments were conducted in a hardware environment equipped with an NVIDIA GeForce RTX 3090 GPU and CUDA version 11.7.

4) Evaluation Metrics: To comprehensively evaluate the quality and diversity of synthetic HSI, we utilized standard metrics commonly used for image generation assessment, including Fréchet inception distance (FID) [83], precision, and recall [84]. FID measures the statistical similarity between the distributions of synthetic and real images, while Precision and Recall quantify the fidelity and diversity of the generated samples, respectively. In the classification experiment, we evaluated the model's performance using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. The OA provides a straightforward measure of the model's overall classification performance across all classes. The AA calculates the mean of the per-class accuracies, offering a more balanced evaluation by treating all classes equally, regardless of class imbalance. The kappa coefficient offers a more precise assessment of the consistency and reliability of the classification results by accounting for random effects.

B. Performance Comparison

1) Classification Performance: To evaluate the effectiveness of FedDA-HSI in mitigating non-IID issues and improving classification performance in FL, we conducted experiments on two widely used hyperspectral land cover datasets: Xiongan and Houston 2018. To simulate statistical heterogeneity across clients, we established an FL setup with four clients. The original HSI was first divided into fixed-size spatial patches, which were then randomly assigned to different clients, resulting in diverse and imbalanced local data distributions. Due to the local correlation of HSI distribution in space, there are still significant differences in data distribution between different clients, even if a random partition is adopted. To evaluate the effectiveness of our proposed method, we compared it with various FL methods, including FedAvg [18], FedALA [60], and FedPAC [61]. In addition, we also assessed the performance of several alternative data augmentation techniques under the same FL framework, including GAN-based augmentation [31], VAE-based augmentation [30], and spectral augmentation [43]. All experiments were conducted under identical federated settings to ensure fairness and reproducibility, with the same number of clients, data partition strategies, class distributions, and communication rounds.

Table I presents the classification results in terms of OA, AA, and the Kappa coefficient, for both the global model and each individual client. As shown in the table, FedDA-HSI consistently achieves superior performance across all metrics on both the Xiongan and Houston 2018 datasets. This confirms the capability of the proposed class-aware augmentation

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TABLE I

QUANTITATIVE COMPARISON OF CLASSIFICATION PERFORMANCE

Dataset	Model	**FedDA-HSI (Ours)**			FedAvg			FedALA			FedPAC			GAN Aug			VAE Aug			Spectral Aug		
		OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Xiongan	Client 1	**72.66**	**52.48**	**0.68**	72.58	51.29	0.68	70.00	46.20	0.65	72.50	51.25	0.68	70.33	47.83	0.66	71.29	49.61	0.66	72.63	51.42	0.68
	Client 2	**71.90**	**48.80**	**0.67**	70.80	47.71	0.66	70.91	47.82	0.66	70.60	47.77	0.66	70.67	45.76	0.66	70.24	47.07	0.65	70.91	47.84	0.67
	Client 3	71.35	49.62	0.67	70.61	45.59	0.66	69.52	49.32	0.64	70.64	45.99	0.66	68.56	49.64	0.63	**71.39**	**51.90**	**0.67**	70.14	44.07	0.65
	Client 4	**72.98**	**54.40**	**0.69**	71.41	50.61	0.66	71.88	50.66	0.66	70.71	50.70	0.65	71.21	45.32	0.64	70.78	46.31	0.66	71.51	50.95	0.67
	Global	**75.61**	**52.32**	**0.71**	74.99	50.96	0.71	75.13	51.24	0.70	74.71	49.41	0.69	74.58	48.46	0.70	74.68	49.43	0.70	75.12	51.44	0.71
Houston 2018	Client 1	**58.60**	**24.25**	**0.45**	57.74	23.26	0.42	57.81	23.37	0.43	57.93	23.53	0.44	58.15	22.08	0.39	56.10	22.96	0.40	57.78	23.29	0.41
	Client 2	**60.46**	**24.29**	**0.44**	58.29	22.66	0.41	58.39	22.88	0.41	58.75	22.52	0.40	57.92	21.18	0.39	57.46	21.10	0.38	58.33	22.53	0.40
	Client 3	**59.87**	21.56	**0.41**	58.72	20.70	0.38	58.73	20.72	0.38	59.67	**22.50**	0.41	57.79	19.06	0.36	58.06	20.36	0.37	58.49	20.61	0.38
	Client 4	**59.36**	**22.19**	**0.41**	56.54	19.90	0.34	56.61	20.00	0.34	57.53	21.59	0.38	55.75	17.12	0.31	56.45	18.41	0.31	57.55	21.60	0.38
	Global	**60.42**	**23.16**	**0.44**	58.44	20.99	0.39	58.51	21.04	0.39	58.18	20.73	0.39	59.00	21.69	0.40	58.25	20.28	0.37	58.66	22.13	0.40

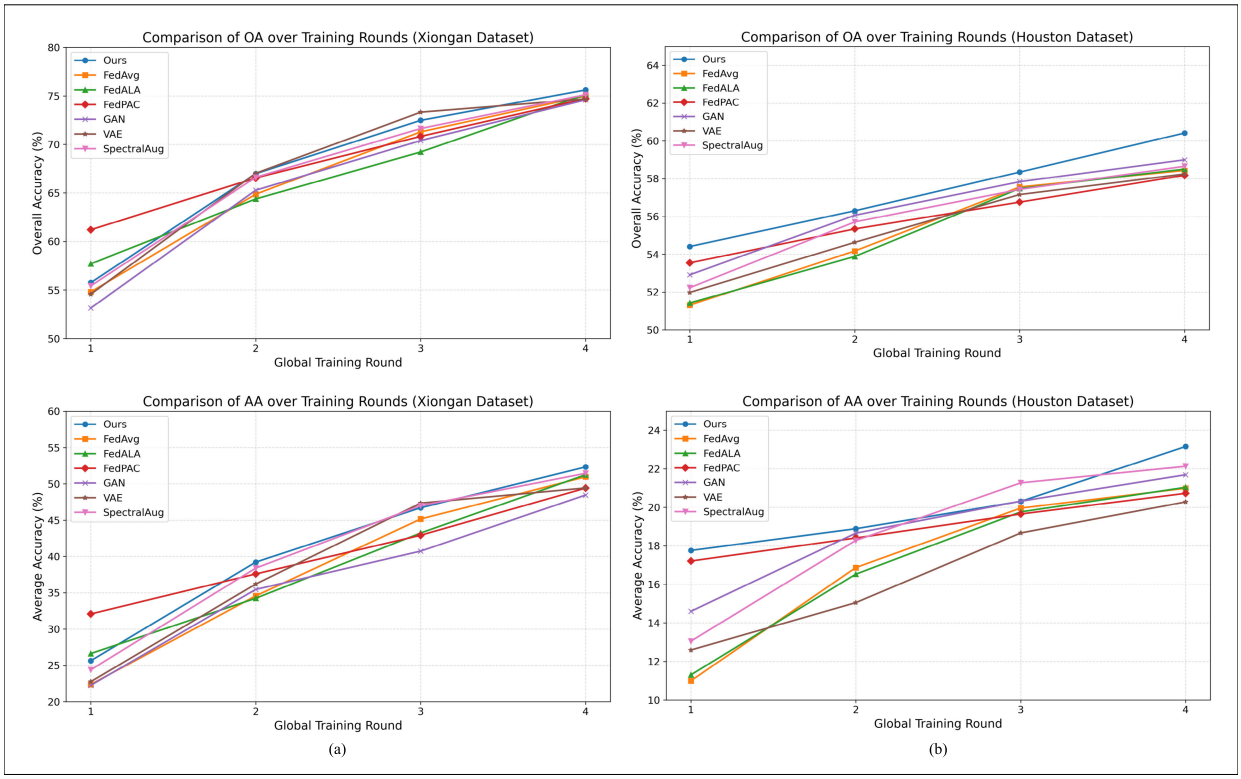


Fig. 3. OA and AA of the global model across training rounds on (a) Xiongan dataset and (b) Houston 2018 dataset.

strategy to enhance model generalization and robustness under non-IID conditions.

Specifically, FedDA-HSI demonstrates significant improvements over the baseline FedAvg algorithm. On the Xiongan dataset, the global model achieves a 0.62 percentage point increase in OA, reaching 75.61%, along with a 1.36 percentage point improvement in AA, and raises the Kappa coefficient to 0.71. Compared to the second-best method, FedALA, our approach still outperforms it by 0.48 percentage points in OA and 1.08 percentage points in AA. Similarly, on the Houston 2018 dataset, the OA increases by 1.98 percentage points, reaching 60.42%, while the AA improves by 2.17 percentage points, and the Kappa coefficient increases to 0.44. Compared to the second-best method, GAN-based augmentation, FedDA-HSI achieves an additional improvement of 1.42 percentage points in OA and 1.47 percentage points in AA. In addition to improving overall classification performance, FedDA-HSI also reduces performance variability among clients, a critical aspect of fairness in FL. For instance, on the Xiongan dataset, the standard deviation of client-level accuracy under FedAvg was 0.769%, which was reduced to 0.638% using FedDA-HSI. Similarly, on the Houston 2018 dataset, the standard deviation decreased from 0.818% to 0.683%. As observed from the performance of individual clients, FedDA-HSI leads to more balanced classification results across clients, effectively mitigating performance disparities in federated settings. To provide a more intuitive comparison, we visualized the OA and AA of the global models for all methods across training rounds. As shown in the accuracy curves in Fig. 3, while performance varied during training, our proposed method ultimately achieves the highest OA and AA among all approaches. The accuracy increases more steadily and reaches a higher final value, indicating improved convergence behavior and enhanced classification performance under non-IID conditions.

To further evaluate the qualitative performance of the proposed FedDA-HSI method, we conducted a visual comparison of classification results across different algorithms on both the Xiongan and Houston 2018 datasets. Specifically, we

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TABLE II
QUANTITATIVE COMPARISON OF SYNTHESIS PERFORMANCE

Dataset	Client 1			Client 2			UnmixDiff(Centralized)		
	FID ↓	Precision ↑	Recall ↑	FID ↓	Precision ↑	Recall ↑	FID ↓	Precision ↑	Recall ↑
Chikusei	54.94	0.64	0.44	55.46	0.64	0.46	58.71	0.64	0.44
Berlin-Urban-Gradient	45.71	0.42	0.68	45.60	0.45	0.73	43.60	0.68	0.53
Xiongan	45.81	0.65	0.58	46.43	0.73	0.62	41.28	0.71	0.53
Houston 2018	67.49	0.70	0.72	69.50	0.79	0.80	68.29	0.80	0.77

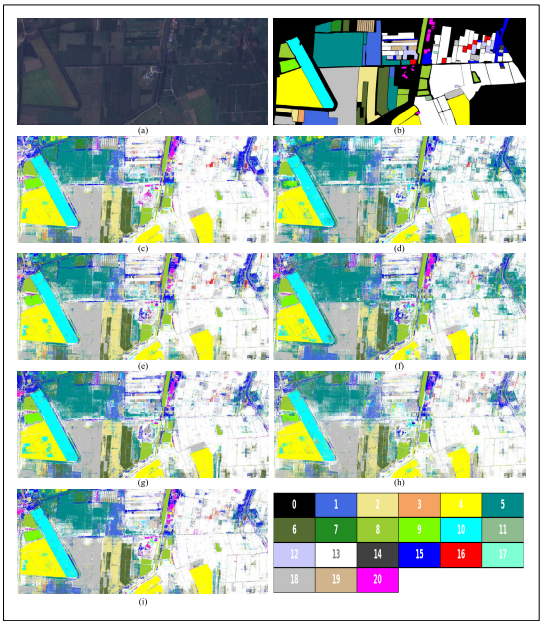


Fig. 4. Classification maps obtained by different methods on the Xiongan dataset. (a) False-color image. (b) GT. (c) FedDA-HSI (Ours). (d) FedAvg. (e) FedALA. (f) FedPAC. (g) GAN Aug. (h) VAE Aug. (i) Spectral Aug.

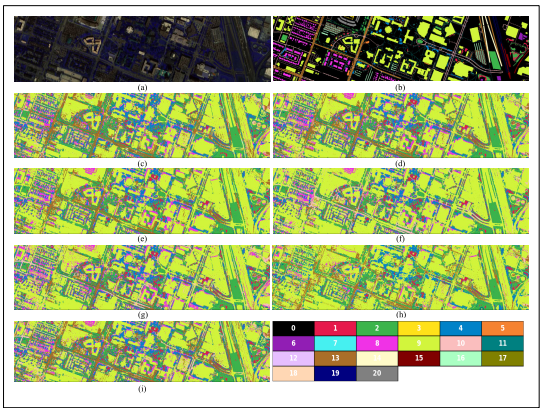


Fig. 5. Classification maps obtained by different methods on the Houston 2018 dataset. (a) False-color image. (b) GT. (c) FedDA-HSI (Ours). (d) FedAvg. (e) FedALA. (f) FedPAC. (g) GAN Aug. (h) VAE Aug. (i) Spectral Aug.

generated pseudo-color HSIs, ground truth maps, and predicted classification maps for each method, as illustrated in Figs. 4 and 5. These visualizations facilitate a more comprehensive comparison of different methods in terms of spatial consistency and overall scene-level classification performance. The results indicate that FedDA-HSI consistently produces more coherent and accurate classification outputs across the entire image, with significantly fewer noisy or misclassified regions. Compared to other approaches, the predictions generated by FedDA-HSI exhibit improved alignment with the underlying ground truth distribution and demonstrate clearer and more well-defined category boundaries. For example, on the Xiongan dataset, categories 5 and 19 are better distinguished using our method, while on the Houston 2018 dataset, categories 4 and 20 are more clearly identified. These visual results further confirm the superiority of FedDA-HSI in learning robust and generalizable feature representations from non-IID data. This advantage is manifested not only in the quantitative classification metrics but also in the visual quality of the classification outputs.

2) Synthesis Performance: To evaluate the synthesis performance of the federated diffusion model, we conducted experiments on four hyperspectral datasets, covering both unconditional and conditional generative settings. For comparative analysis, we established baseline performance using centralized training on each dataset. The evaluation metrics were computed between the generated HSIs and the real HSIs, and the results are summarized in Table II. The results demonstrate that our proposed method is capable of generating diverse, high-quality HSIs across all evaluated datasets. Notably, despite operating under non-IID data distributions and strict data privacy constraints, the FL approach achieved generation performance comparable to its centralized counterpart. This highlights the potential of federated diffusion models to achieve strong generative performance while preserving data locality and addressing real-world distributional challenges.

C. Ablation Study on Augmentation Strategies

To validate the effectiveness of the proposed class-aware augmentation strategy, we conducted an ablation study comparing different augmentation settings within the same FL framework. The first setting is the baseline (no augmentation), where the global classifier is trained solely on the original local data without incorporating any synthetic samples. The second setting, termed per-round conditional augmentation, involves each client augmenting its local dataset during every global training round by generating class-conditional hyperspectral samples using the current aggregated global diffusion model. The third setting corresponds to our proposed class-aware augmentation, in which synthetic samples are generated only once after the final global training round.

We present the performance comparison of the three augmentation strategies in Table III. Compared to the baseline, both augmentation strategies lead to improvements, confirming

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TABLE III
ABLATION STUDY ON AUGMENTATION STRATEGIES

Dataset	Method	Client 1			Client 2			Global		
		OA ↑	AA ↑	Kappa ↑	OA ↑	AA ↑	Kappa ↑	OA ↑	AA ↑	Kappa ↑
Xiongan	Baseline (No Aug)	80.28	60.47	0.77	81.59	64.74	0.79	80.77	65.16	0.77
	Per-Round Aug	80.51	60.62	0.76	81.50	64.89	0.77	80.92	65.75	0.78
	Class-Aware Aug (Ours)	**80.94**	**61.34**	**0.78**	**81.75**	**66.35**	**0.79**	**82.69**	**66.62**	**0.78**
Houston 2018	Baseline (No Aug)	71.47	53.70	0.62	70.33	41.11	0.60	72.84	47.89	0.63
	Per-Round Aug	71.49	53.80	0.61	70.53	42.60	0.60	72.92	47.98	0.63
	Class-Aware Aug (Ours)	**71.74**	**54.91**	**0.62**	**71.18**	**44.64**	**0.61**	**73.18**	**48.21**	**0.63**

that data generation positively enhances model generalizability in federated settings. However, the per-round conditional augmentation approach yields suboptimal performance and exhibits considerable variability across clients. Specifically, on the Xiongan dataset, the final global model trained with the per-round conditional augmentation strategy achieved an OA improvement of 0.15% and an AA improvement of 0.59% over the baseline, with a Kappa coefficient of 0.78. In comparison, our class-aware augmentation strategy led to more substantial improvements, increasing the OA by 1.92% and AA by 1.46%, while achieving a Kappa coefficient of 0.78. For the Houston 2018 dataset, the per-round conditional augmentation approach improved the OA by 0.08% and AA by 0.09% compared to the baseline, with a Kappa coefficient of 0.63. Our class-aware augmentation strategy demonstrated superior performance with a 0.34% OA improvement and 0.32% AA improvement, while achieving a Kappa coefficient of 0.63. This is primarily due to the instability of the generative model in early global training rounds, during which low-quality samples are produced. These early stage synthetic samples offer limited benefit, or even introduce noise, to model training. Consequently, performance improvements become apparent only in the later stages. In addition, the continual generation of new data in each round introduces uncertainty, which may amplify local distribution bias and result in greater interclient performance disparity. In contrast, our proposed Class-Aware Augmentation strategy generates high-quality, semantically controlled samples using the final converged global diffusion model. This enables more stable and efficient enhancement, improving both classification accuracy and client-level fairness under non-IID conditions.

D. DP Experiment

To assess the effectiveness of our proposed DP strategy, we conducted experiments comparing the performance of DP-enhanced generation models against standard models without DP. Our aim was to quantify the tradeoff between privacy and utility, specifically focusing on classification accuracy.

The experimental results, summarized in Table IV, present the performance metrics of clients using DP, compared to those without it, as well as the FedAvg results. The results indicate that introducing DP leads to only a slight decrease in classification metrics such as OA and AA at the client level. Importantly, despite this minor drop, the accuracy achieved with our DP approach surpasses that of the traditional FedAvg model. This demonstrates that our DP strategy effectively preserves classification performance while enhancing privacy protection, successfully balancing the tradeoff between accuracy and privacy.

TABLE IV
PERFORMANCE COMPARISON OF CLIENTS WITH AND WITHOUT DP

Client ID	Differential Privacy			*w/o* Differential Privacy			FedAvg		
	OA ↑	AA ↑	Kappa ↑	OA ↑	AA ↑	Kappa ↑	OA ↑	AA ↑	Kappa ↑
Client 1	72.37	48.99	0.68	72.66	52.48	0.68	72.58	51.29	0.68
Client 2	71.01	44.75	0.66	71.90	48.80	0.67	70.80	47.71	0.66
Client 3	72.74	53.14	0.68	71.35	49.62	0.67	70.61	45.59	0.65
Client 4	72.60	48.57	0.68	72.98	54.50	0.69	71.41	50.61	0.66
Global	75.53	50.00	0.71	75.61	52.32	0.71	74.99	50.96	0.71

TABLE V
EVALUATION OF GENERATION QUALITY UNDER VARYING CLIENT NUMBERS

Client Number	Client ID	FID ↓	Precision ↑	Recall ↑
k=2	Client 1	45.81	0.65	0.58
	Client 2	46.43	0.73	0.62
k=4	Client 1	61.78	0.66	0.48
	Client 2	62.74	0.67	0.50
	Client 3	63.41	0.63	0.40
	Client 4	61.80	0.72	0.51
k=6	Client 1	67.28	0.63	0.46
	Client 2	67.43	0.60	0.47
	Client 3	70.11	0.56	0.52
	Client 4	65.65	0.69	0.44
	Client 5	65.97	0.63	0.53
	Client 6	62.02	0.70	0.63
k=8	Client 1	75.01	0.65	0.62
	Client 2	81.89	0.63	0.67
	Client 3	78.40	0.76	0.61
	Client 4	80.81	0.54	0.60
	Client 5	81.17	0.52	0.55
	Client 6	74.98	0.49	0.72
	Client 7	84.51	0.65	0.59
	Client 8	66.15	0.70	0.69

E. Parameter Sensitivity Analysis

1) Impact of Client Numbers: To evaluate the robustness of the FedDA-HSI under varying federation scales, we conducted experiments with 2, 4, 6, and 8 clients. Tables V and VI show the performance of the generative and classification models under these four configurations. For the generative model, our results reveal a correlation between client numbers and model performance. We observed that the FID score gradually increased as the number of clients increased from 2 to 8. With fewer clients, the model produces higher quality samples, demonstrating efficient knowledge aggregation

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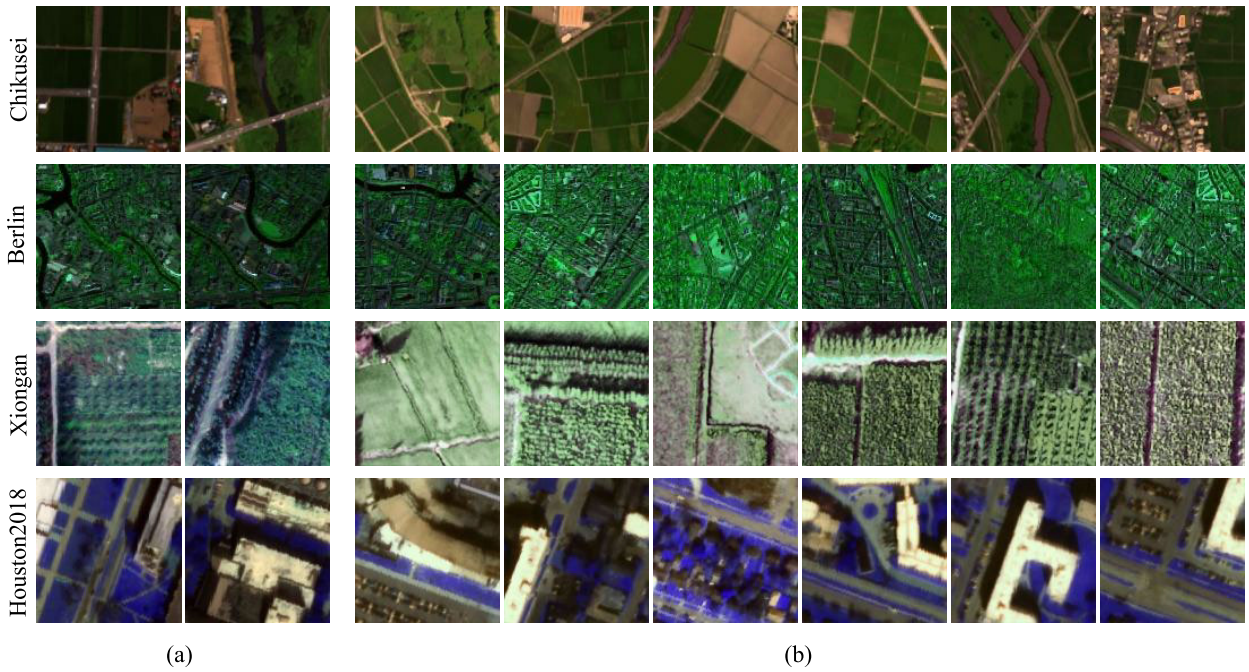


Fig. 6. Visual comparison of synthesis results. (a) Original HSI serving as reference. (b) Synthetic images generated by our model.

TABLE VI
COMPARISON OF DIFFERENT CLIENT NUMBERS

Client Num	Model	Ours			*w/o* Augmentation		
		OA ↑	AA ↑	Kappa ↑	OA ↑	AA ↑	Kappa ↑
k=2	Client 1	**80.94**	**61.34**	**0.78**	80.28	60.47	0.77
	Client 2	**81.75**	**66.35**	**0.79**	81.59	64.74	0.79
	Global	**82.69**	**66.62**	**0.78**	80.77	65.16	0.77
k=4	Client 1	**72.66**	**52.48**	**0.68**	72.58	51.29	0.68
	Client 2	**71.90**	**48.80**	**0.67**	70.80	47.71	0.66
	Client 3	**71.35**	**49.62**	**0.67**	70.61	45.59	0.66
	Client 4	**72.98**	**54.40**	**0.69**	71.41	50.61	0.66
	Global	**75.61**	**52.32**	**0.71**	74.99	50.96	0.71
k=6	Client 1	**62.19**	**34.97**	**0.56**	60.53	33.86	0.54
	Client 2	63.37	**44.17**	**0.57**	**63.52**	40.13	0.57
	Client 3	**63.56**	39.44	**0.57**	63.41	**39.93**	0.57
	Client 4	**62.65**	**33.13**	**0.56**	62.29	31.91	0.55
	Client 5	60.77	**40.56**	**0.55**	**61.62**	37.86	0.55
	Client 6	**61.93**	**35.77**	**0.55**	61.58	34.74	0.54
	Global	**67.00**	**39.11**	**0.61**	63.38	31.48	0.57
k=8	Client 1	**61.23**	**39.86**	**0.54**	59.68	37.70	0.53
	Client 2	62.52	**39.34**	**0.56**	**62.95**	36.95	0.56
	Client 3	**61.93**	**36.91**	**0.56**	61.72	36.11	0.55
	Client 4	**61.92**	**35.28**	**0.55**	59.71	32.72	0.53
	Client 5	**60.73**	**36.20**	**0.55**	59.80	32.49	0.53
	Client 6	**61.04**	**36.06**	**0.54**	60.00	32.03	0.53
	Client 7	61.27	**39.64**	0.54	**61.78**	38.76	**0.55**
	Client 8	**62.29**	**37.06**	**0.56**	61.02	34.65	0.55
	Global	**68.12**	**38.07**	**0.62**	65.90	34.94	0.59

in simpler federated scenarios. Although performance shows a slight decline as the number of clients increases to 8, the generated samples maintain acceptable quality standards. This trend indicates that increased federated heterogeneity introduces greater challenges for the generative model, yet the decline remains smooth and controlled, reinforcing the robustness of FedDA-HSI across varying scales.

In the classification experiments, FedDA-HSI consistently outperforms the benchmark approach without data augmentation across different client configurations, highlighting the effectiveness of our data augmentation strategy. The advantage becomes particularly evident as the number of clients increases. When the number of clients is 2, FedDA-HSI surpasses the baseline in terms of global model performance, improving the OA by 1.92 percentage points, from 80.77% to 82.69%. The AA also increases by 1.46 percentage points, and the Kappa coefficient reaches 0.78. For the case of four clients, FedDA-HSI shows a notable improvement, with OA increasing by 0.62 percentage points, resulting in a new performance metric of approximately 75.61%. The AA improves by 1.36 percentage points, and the Kappa coefficient reaches 0.71. With six clients, the performance gains are the most pronounced, as FedDA-HSI achieves an OA increase of 3.62 percentage points, significantly surpassing the baseline. The AA shows remarkable improvement as well, rising by 7.63 percentage points, while the Kappa coefficient stabilizes at 0.61, indicating a reliable enhancement in model performance as client numbers grow. In the case of eight clients, FedDA-HSI still maintains competitive performance, outperforming the baseline by 2.22 percentage points in OA, increasing it from 65.90% to 68.12%. The AA improves by 3.13 percentage points, and the Kappa coefficient reaches 0.62. While the benchmark method shows substantial accuracy variations among clients due to greater data distribution disparities, FedDA-HSI maintains minimal accuracy differences between clients. Under the eight-client setting, the standard deviation is reduced from 1.149 to 0.597, suggesting better robustness to data heterogeneity. This balanced performance validates that our data enhancement strategy effectively addresses the data heterogeneity challenge in FL, while promoting fairness across clients. These results demonstrate that FedDA-HSI successfully combines robust generation capabilities with effective data distribution balancing, providing a practical solution for non-IID scenarios in FL systems. The consistent

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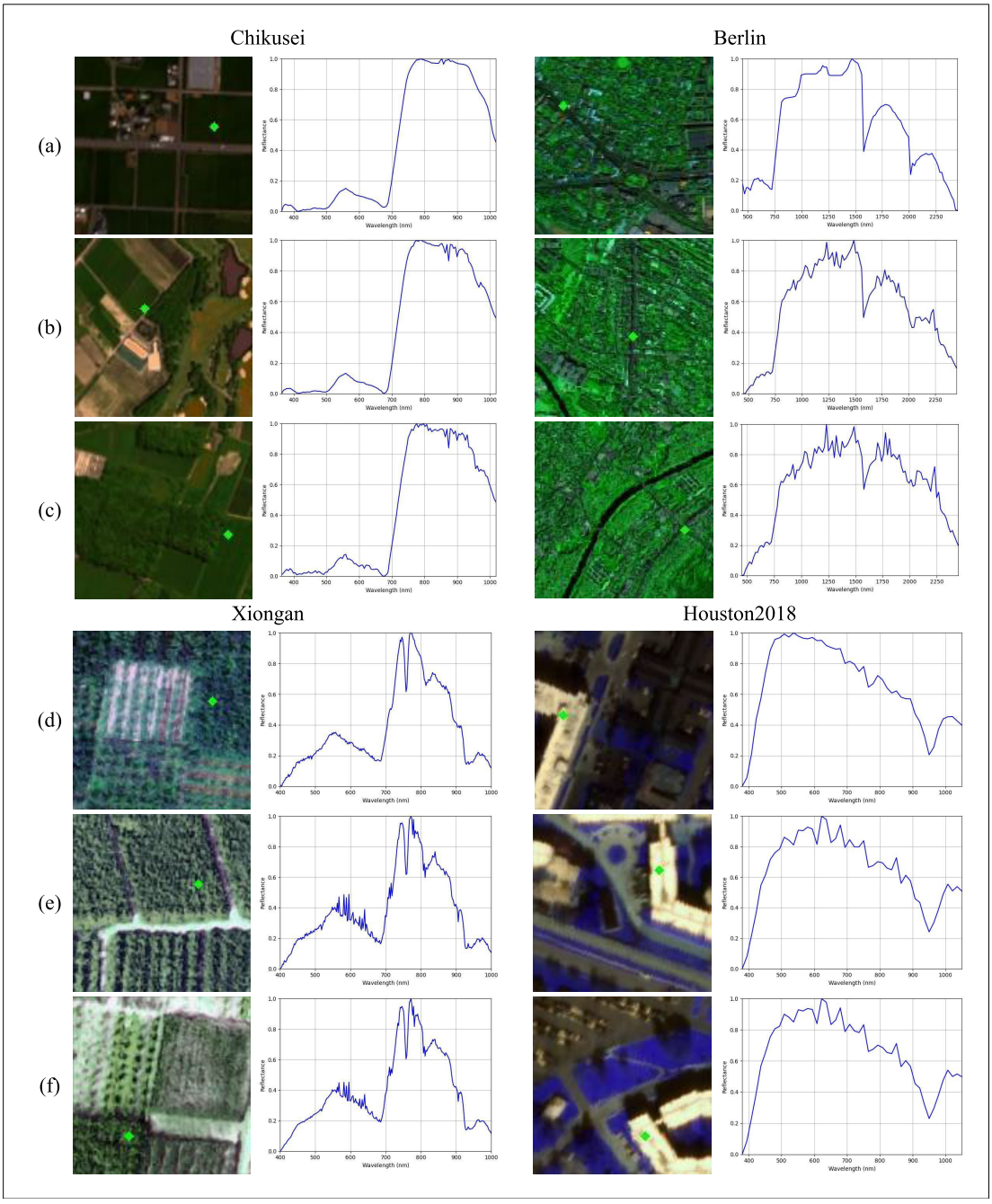


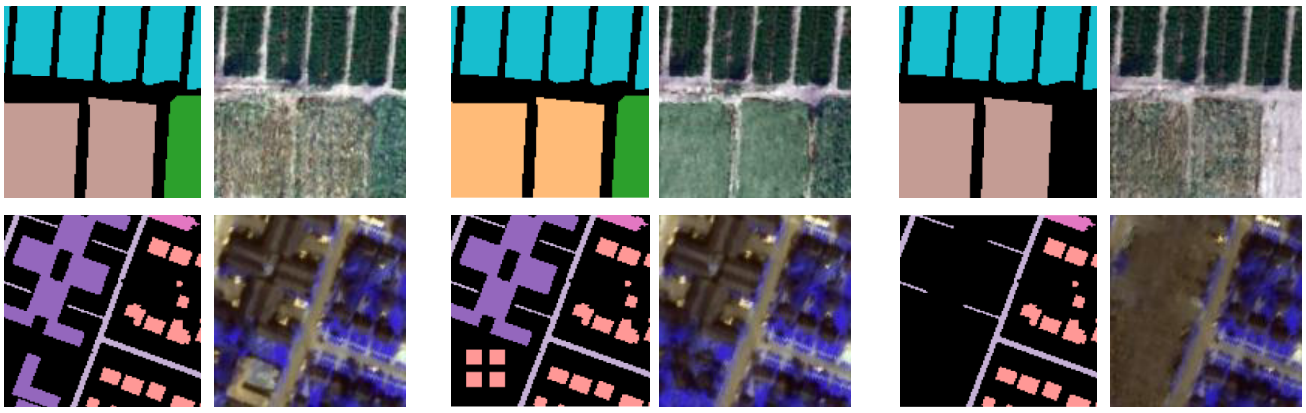
Fig. 7. Spectral profile comparison between real and generated HSIs. (a) and (d) Original data with sampled spectrum. (b), (c), (e), and (f) Generated results with corresponding spectral spectrum. Sampling positions marked in green. Plots show reflectance across wavelength bands.

performance across different client numbers further confirms the scalability and adaptability of our approach in real-world FL applications.

2) Impact of Data Splitting: To further investigate the impact of varying degrees of data heterogeneity among clients, we conducted additional experiments using the Xiongan dataset by applying a Dirichlet distribution-based partitioning strategy with different concentration parameters $\alpha \in \{0.5, 0.3, 0.1\}$. The Dirichlet distribution is widely adopted in FL to simulate realistic non-IID data scenarios, where smaller values of α correspond to higher degrees of label imbalance and client divergence.

Table VII summarizes the classification results under different Dirichlet concentration parameters. As expected, the overall model performance shows a gradual decline as data heterogeneity increases (i.e., as α decreases), primarily due to the growing divergence among client data distributions and reduced class coverage within individual clients. Nevertheless, even in the highly skewed scenario of $\alpha = 0.1$, our proposed framework maintains strong performance, demonstrating robust generalization and resilience to severe non-IID conditions. Moreover, the incorporated class-aware generation strategy effectively mitigates performance degradation caused by local class scarcity during training, significantly

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(a) (b) (c)

Fig. 8. Conditional generation results using semantic labels. (a) Initial semantic label and corresponding generated image. (b) Generation results with region modification. (c) Generation results with region removal.

TABLE VII

CLASSIFICATION PERFORMANCE ON THE XIONGAN DATASET UNDER VARYING DIRICHLET α VALUES ACROSS FOUR CLIENTS

Distribution	Model	Ours			*w/o* Augmentation		
		OA ↑	AA ↑	Kappa ↑	OA ↑	AA ↑	Kappa ↑
$\alpha = 0.5$	Client 1	**72.66**	**52.48**	**0.68**	72.58	51.29	0.68
	Client 2	**71.90**	**48.80**	**0.67**	70.80	47.71	0.66
	Client 3	**71.35**	**49.62**	**0.67**	70.61	45.59	0.66
	Client 4	**72.98**	**54.40**	**0.69**	71.41	50.61	0.66
	Global	**75.61**	**52.32**	**0.71**	74.99	50.96	0.71
$\alpha = 0.3$	Client 1	**68.11**	**45.49**	**0.62**	67.84	44.13	0.62
	Client 2	**71.26**	**48.63**	**0.66**	70.48	47.00	0.65
	Client 3	**66.14**	**43.88**	**0.60**	65.66	42.15	0.59
	Client 4	**77.49**	**56.09**	**0.74**	75.97	53.40	0.72
	Global	**77.05**	**54.23**	**0.73**	75.45	52.12	0.71
$\alpha = 0.1$	Client 1	**64.14**	**35.95**	**0.58**	63.71	33.72	0.57
	Client 2	**72.58**	**49.69**	**0.68**	71.40	48.81	0.67
	Client 3	**74.89**	**53.36**	**0.71**	73.51	51.90	0.69
	Client 4	**70.95**	**47.01**	**0.66**	70.17	45.60	0.65
	Global	**75.52**	49.50	**0.71**	73.62	**49.66**	0.69

enhancing the model's ability to recognize minority classes. These experimental results collectively verify the adaptability and stability of our approach when confronted with challenging heterogeneous data distributions.

F. Comparison With Centralized SOTA on Houston 2018 Dataset

To further validate the effectiveness of FedDA-HSI, we conducted experiments on the Houston 2018 dataset and compared our method with the recent centralized method GraphGST [85], using the official implementation. While GraphGST assumes full access to centralized data, which is often unrealistic in privacy-sensitive scenarios, our method operates under federated settings without data sharing.

Despite this difference, FedDA-HSI achieves competitive performance, showing strong classification results even under non-IID and decentralized conditions. Table VIII presents the comparison, demonstrating that our framework provides a practical and privacy-preserving alternative to centralized approaches like GraphGST.

TABLE VIII

PERFORMANCE COMPARISON BETWEEN OUR FEDERATED METHOD AND CENTRALIZED GRAPHGST ON THE HOUSTON 2018 DATASET

Method	OA ↑	AA ↑	Kappa ↑
Ours (Federated)	73.18	48.21	0.63
GraphGST (Centralized)	80.93	89.26	0.76

TABLE IX

MODEL PARAMETERS, FLOPS, AND INFERENCE TIME OF FEDDA-HSI EVALUATED ON TWO HYPERSPECTRAL DATASETS

Dataset	para.	FLOPs	Time
Xiongan	100.17M	0.21M	0.26ms
Houston 2018	100.07M	0.05M	0.23ms

G. Computational Complexity and Efficiency

To assess the computational cost and practical feasibility of the proposed FedDA-HSI framework, Table IX presents the number of parameters, FLOPs per image, and average inference time on the Xiongan and Houston 2018 datasets. The results indicate that our model contains approximately 100 million parameters, which encompass the classifier module as well as latent representations derived from the diffusion and unmixing models during the training phase. Nonetheless, the actual inference cost remains minimal, since only the classifier is involved during prediction. In particular, the FLOPs per image are very low, with 0.21 M on Xiongan and 0.05 M on Houston 2018, and the average inference time stays below 0.3 ms per image. This makes the model suitable for semi-real-time or offline remote sensing applications.

Although the overall model framework includes a diffusion-based generative component and a spectral unmixing module, these components are solely used during local data augmentation at the training stage and do not affect the computational burden during inference. This decoupled design ensures computational efficiency while allowing the framework to generate robust, class-balanced datasets that improve generalization performance under challenging non-IID conditions.

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H. Result Visualization

We assessed the federated diffusion model on four benchmark HSI datasets. Visually, the synthetic patches match real imagery in both fine texture and large-scale layout (Fig. 6). Spectrally, class-specific reflectance curves of farmland, vegetation, roofs, and roads align closely with ground truth, confirming accurate band-wise reconstruction (Fig. 7). A semantic-control test shows the model can add or remove target classes on demand (Fig. 8), demonstrating reliable conditional generation. These results verify that our approach preserves spatial detail, spectral fidelity, and editing flexibility while maintaining data privacy.

V. Conclusion

We present FedDA-HSI for HSI classification that overcomes the two main obstacles of label scarcity and non-IID data. A client-side diffusion model produces high-fidelity synthetic spectra without exposing raw imagery, achieving spatial and spectral quality on par with centralized training across four benchmarks. Our class-aware augmentation further balances label distributions by generating samples for missing or rare classes, boosting both accuracy and interclient fairness across different federation scales. These results establish a strong baseline for secure, collaborative learning in remote sensing. While effective across diverse settings, the proposed framework assumes a reasonable amount of local data per client to train the diffusion model and support class-aware generation, which may limit its effectiveness when applied to extremely small-scale datasets. Future work will extend the method to more extreme non-IID settings, design adaptive augmentation policies, and integrate stronger privacy guarantees.

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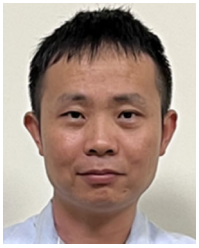
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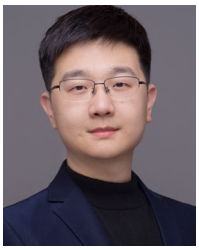
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